Ethics and Bias in AI

Assignment Reference Guide

A content-rich revision and application guide focused on ethical problem decisions, VSD, stakeholder analysis, documentation, audits, and recommendations.

Purpose of this guide

- Turn the lecture material into an **assignment-ready reference** rather than a generic summary.
- Keep the **terminology, frameworks, and theory** intact while making them easy to reuse in your presentation and justification.
- Prioritise the content most likely to help with: **ethical problem framing, stakeholder analysis, value tensions, empirical audit interpretation, documentation critique, and actionable recommendations**.

Slides this guide draws on

- Week 1: introduction to ethics and bias
- Week 2: societal impacts of AI
- Week 3: bias and fairness
- Week 4/5: Value Sensitive Design and its applications to AI
- Week 6: audits and evaluation
- Week 8: transparency and accountability
- Lecture 10: harm and bias mitigation

Prepared as a study aid from the uploaded lecture slides. Useful for structuring arguments, but still pair it with your own audit findings and critical judgement.

Contents

1 What this assignment is really asking you to do **3**
1.1 High-mark pattern to follow . . . 3
1.2 Common weak moves to avoid . . . 3

2 Ethics foundations you can explicitly use **3**
2.1 Morality, ethics, and AI ethics . . . 3
2.2 Three ethical theories you can use to justify decisions . . . 4
2.3 Ethical principles and frameworks worth naming . . . 4

3 Societal harms and ethical issues you should be alert to **5**
3.1 Recurring harms from the slides . . . 5
3.2 Human bias matters because AI systems are built by humans . . . 5
3.3 Protected characteristics . . . 5

4 Bias and fairness: the theory you are most likely to apply **6**
4.1 Two meanings of "bias" . . . 6
4.2 Types of bias across the ML pipeline . . . 6
4.3 Especially relevant to your topic: annotation bias in toxicity detection . . . 7
4.4 Fairness is not one thing . . . 7
4.5 Common group fairness definitions . . . 7
4.6 Fairness through unawareness is not enough . . . 7
4.7 Confusion matrix and error metrics . . . 8
4.8 How to talk about fairness well . . . 8

5 Stakeholder analysis: make it precise, not generic **8**
5.1 Stakeholder definitions . . . 8
5.2 Direct and indirect stakeholders . . . 8
5.3 A practical stakeholder map for a content moderation assignment . . . 9

6 Value Sensitive Design (VSD): the key design framework **9**
6.1 Definition and stance . . . 10
6.2 Tripartite structure . . . 10
6.3 Seven-step VSD process . . . 10
6.4 Useful value vocabulary from the slides . . . 10
6.5 Value tensions and designer values . . . 11
6.6 From abstract values to concrete design . . . 11
6.7 VSD applied to AI / ML . . . 11

7 Transparency, documentation, and accountability **11**
7.1 Why documentation matters . . . 12
7.2 Data Statements . . . 12
7.3 Other documentation practices you can name . . . 12
7.4 Questions to ask when critiquing documentation . . . 12
7.5 Benefits and barriers . . . 13

8 Audits and evaluation **13**
8.1 What an algorithmic audit is . . . 13
8.2 Goals of an audit . . . 13
8.3 How to design an audit . . . 13
8.4 Internal vs external audits . . . 14
8.5 Important audit cautions from the slides . . . 14

9 From evidence to recommendations **14**
9.1 Recommendation categories that fit this module well . . . 14
9.2 What makes a recommendation strong . . . 15
9.3 Recommendation templates you can adapt . . . 15

10 A fast workflow for your own ethical problem decisions **16**

11 Compact glossary of high-value terms **16**

12 Final takeaways to remember in your presentation **18**

What this assignment is really asking you to do

Core assessment logic
A strong submission does not just identify that "bias exists." It should do five things together:

1. define a **specific ethical problem** rather than a vague concern;
2. identify a **focused stakeholder set** and explain why those stakeholders matter;
3. connect findings from your **empirical audit** to the underlying design, data, and sociotechnical issues;
4. critically assess the **data documentation / transparency material**;
5. produce **actionable recommendations** tied to values, harms, trade-offs, and evidence.

High-mark pattern to follow

A reusable sentence structure
Claim → Evidence → Who is affected → Which value is implicated → What tension exists → What should be changed and why
Example pattern:

> "The dataset / classifier raises a **representation or annotation problem**. This matters because it produces **differential risk** for **stakeholder X**. The value tension is between **A** and **B**. Therefore, a strong recommendation is **intervention Y**, supported by **audit/documentation evidence Z**."

Common weak moves to avoid

- Saying a system is "biased" without specifying **which type of bias**, **for whom**, and **how you know**.
- Treating fairness as one number rather than a **choice among competing definitions**.
- Listing stakeholders but not explaining their **stakes, values, or harms**.
- Proposing recommendations that are morally attractive but **too vague to implement**.
- Ignoring the **sociotechnical setting**: annotators, deployment context, users, incentives, governance, and feedback loops.

Ethics foundations you can explicitly use

Morality, ethics, and AI ethics

Term	Meaning	Why it helps in the assignment
Morality	Actions are moral if they are "good" or worthy of praise.	Helps distinguish the *moral judgement* from the *framework* used to justify it.
Ethics	Formal study of moral standards and conduct; aims to construct a general basis for deciding what is moral.	Lets you show that your recommendations are **principled**, not just intuitive.
AI ethics	Values, principles, and techniques used to guide moral conduct in the development and use of AI technologies.	Useful when moving from abstract ethics to **system design and governance**.
Data ethics	Moral problems related to data generation, recording, curation, processing, dissemination, sharing, use, algorithms, and related practices.	Very useful for a **dataset-focused assignment**, because it foregrounds curation and documentation rather than only the model.

Three ethical theories you can use to justify decisions

Theory	Core question	How to use it in discussion
Virtue ethics	What is the good / admirable thing to do?	Good for arguing that system design should cultivate integrity, care, honesty, and responsible professional conduct.
Deontological ethics	Are we following the right rules, duties, and constraints?	Good for privacy, consent, rights, transparency duties, protected characteristics, and non-discrimination obligations.
Consequentialist ethics	What outcomes follow from this design choice?	Good for comparing harms, trade-offs, false positives, false negatives, chilling effects, and downstream social impacts.

Ethical principles and frameworks worth naming

- **EU Guidelines on Trustworthy AI**: human agency and oversight; technical robustness and safety; privacy and data governance; transparency; diversity, non-discrimination and fairness; societal and environmental well-being; accountability.
- **ACM Code of Ethics and Professional Conduct**: useful for framing professional responsibilities and the duty to benefit society.
- **AI for human flourishing / AI for good**: contrasts socially desirable purposes with efficiency, optimisation, profit maximisation, or social control.

Assignment tip
When you justify a recommendation, it is often stronger to say *which principle is being protected*. Example: "This recommendation strengthens accountability and transparency, while also supporting fairness and human oversight."

Societal harms and ethical issues you should be alert to

Recurring harms from the slides

- environmental impacts
- value alignment issues
- harmful content and inaccuracies
- privacy issues and data breaches
- exploitation of workers
- copyright infringement
- labour-force / sector transformation
- unequal access to AI as costs evolve
- impacts on social and individual well-being
- opacity and lack of appeal
- feedback loops
- over-trust in "objective" systems

Human bias matters because AI systems are built by humans

- **Social biases**: snap judgements about people based on social group membership.
- **Cognitive biases**: systematic tendencies in thought that lead to error, such as **confirmation bias**.
- **Exposure effect**: repeated claims can become familiar and therefore more believable.
- **Implicit bias**: automatic associations that can produce unintentionally differential or unfair treatment.

Protected characteristics

The slides explicitly connect bias and discrimination to protected attributes / characteristics. The Equality Act 2010 list mentioned in the slides includes:

- age
- race
- sex
- sexual orientation
- disability
- marriage and civil partnership
- gender reassignment
- religion or belief
- maternity and pregnancy

Why this matters for content moderation
For a toxicity or hate-speech dataset, the ethical problem may not only be "model accuracy." It may concern **who is over-flagged**, **whose speech is misread**, **whose safety is under-protected**, and **which social groups bear the costs of moderation errors**.

Bias and fairness: the theory you are most likely to apply

Two meanings of "bias"

- In statistics / machine learning, bias can simply mean an inductive preference or basis for selecting one generalisation over another.
- In this module, the key concern is **algorithmic bias as unfair or unjust discrimination** against individuals or groups, especially where this maps onto historically marginalised characteristics.

Types of bias across the ML pipeline

Bias type	Meaning	What to ask in an assignment
Historical bias	The world that generated the data is itself unjust, outdated, or misaligned with the values the system should promote.	Does the dataset reproduce past discrimination, outdated social roles, or harmful norms?
Representation bias	The sampled development population under-represents part of the use population.	Who is missing, under-sampled, or poorly represented?
Measurement bias	Features / labels are chosen or measured badly, often via poor proxies.	Are labels valid? Are you measuring the thing you claim to measure?
Annotation bias	Human labelling embeds subjective or socially patterned judgements.	Who annotated the data, under what instructions, and with what context?
Aggregation bias	Distinct populations are combined even when a single model does not fit all subgroups.	Are subgroup differences being collapsed into a misleading average?
Evaluation bias	Benchmark or test data does not reflect the population the system will serve.	Are the evaluation set and metrics appropriate for the real use case?
Deployment bias	The system is used differently from how it was intended or evaluated.	What happens when the model enters a real institution with human decision-makers and incentives?

Especially relevant to your topic: annotation bias in toxicity detection

The slides highlight that toxicity / hate-speech labelling involves a **subjective assessment of what counts as offensive**. They also note that **African American dialect may be more likely to be labelled as toxic / hate speech**, and that this can be improved by better annotator training and added contextual information.

> **Interpretive move that often earns marks**
> Do not only ask whether a label is correct. Ask whether the **labelling regime itself** reflects narrow linguistic norms, insufficient context, or a mismatch between annotator background and the communities being labelled.

Fairness is not one thing

Fairness concept	Meaning	What it foregrounds
Procedural fairness	Is the process that determines the outcome fair?	rules, consistency, transparency, due process
Outcome fairness	Are the resulting distributions of outcomes fair?	who benefits, who is burdened, distributive justice
Equality	Same treatment / same allocation	parity, sameness
Equity	Different support to account for different needs / conditions	fairness with context, not mere sameness

Common group fairness definitions

- **Demographic parity**: equal positive prediction rates across groups.
- **Equalized odds / equal opportunity**: equal true-positive and/or false-positive rates across groups.
- **Calibration**: the same score should mean the same thing regardless of group.

> **Critical point**
> The slides stress that fairness is **mathematically plural**. There are many definitions and metrics, and they cannot all be satisfied simultaneously. So the real question is not simply technical; it is ethical: **which errors matter more, and who gets to decide?**

Fairness through unawareness is not enough

Removing a protected attribute such as race does *not* guarantee fairness if the model can reconstruct it from correlated variables. In other words, systems may still discriminate via profiling, redundant encodings, or indirect evidence.

Confusion matrix and error metrics

	Outcome = Positive	Outcome = Negative
Prediction = Positive	True Positive (TP)	False Positive (FP)
Prediction = Negative	False Negative (FN)	True Negative (TN)

$$\text{Error Rate} = \frac{FP + FN}{TP + FP + FN + TN}$$

$$\text{False Positive Rate} = \frac{FP}{FP + TN} \qquad \text{False Negative Rate} = \frac{FN}{FN + TP}$$

- False positives often matter to people who are wrongly flagged, penalised, or excluded.
- False negatives often matter to those prioritising risk reduction, enforcement, or safety.
- This is why **different stakeholders will value different metrics differently**.

How to talk about fairness well

- Choose metrics that actually capture the question you care about.
- Present them carefully; audiences can easily misread them.
- No single metric tells the whole story.
- Group fairness metrics usually depend on demographic information, which may be absent or incomplete.
- Group metrics can still miss **intersectionality**.

Stakeholder analysis: make it precise, not generic

Stakeholder definitions

- **Stakeholders**: those who are or will be significantly implicated by a technology.
- Stakeholders can be people, groups, neighbourhoods, communities, organisations, institutions, or societies.
- VSD asks designers to seek a **robust set of stakeholder groups** and provide an **analytical or empirical rationale** for including them.

Direct and indirect stakeholders

Category	Definition
Direct stakeholders	individuals or organisations who interact directly with the system and its outputs
Indirect stakeholders	parties who do not directly interact with the system but are still affected by its use

Two extra points from the slides matter a lot:

- **Roles, not individuals**: the same person or entity may hold multiple relationships to the same technology.
- In AI system design, direct and indirect categories may be **less clear than they first appear**, and stakeholder roles can shift across contexts.

A practical stakeholder map for a content moderation assignment

Stakeholder	Direct / indirect	Main values at stake	Typical harms / concerns
Users who post comments	direct	expression, autonomy, due process, fair treatment	over-flagging, misclassification, dialect bias, chilling effects
Targets of abuse / harassment	indirect (sometimes direct)	safety, dignity, freedom from harm	under-moderation, delayed action, retraumatisation
Human moderators	direct	well-being, safety, workload fairness, accountability	exposure to harmful content, burnout, unfair reliance on automation
Platform trust & safety teams	direct	consistency, risk management, accountability	poorly scoped metrics, over-trust in model outputs, governance failures
Annotators / data workers	direct or upstream	fair labour conditions, contextual understanding, recognition of expertise	invisibilised labour, poorly specified labelling tasks
Marginalised language communities	indirect	fairness, inclusion, non-discrimination, recognition of linguistic variation	disproportionate flagging, stereotyping, exclusion
Researchers / downstream reusers	indirect	transparency, valid documentation, scientific quality	misuse through bad generalisation assumptions
Regulators / civil society groups	indirect	accountability, rights protection, public trust	lack of transparency, no appeal path, no evidence base

What makes stakeholder analysis strong?

- not just naming stakeholders, but explaining **why they have a lot at stake**;
- identifying **benefits and harms** for each group;
- mapping those benefits and harms onto **values**;
- showing where values come into **tension**.

Value Sensitive Design (VSD): the key design framework

Definition and stance

Value Sensitive Design (VSD) is a design approach that attempts to systematically include values of ethical importance in design. It assumes that technology design bears directly and systematically on the realisation or suppression of social, ethical, and political values.

Key VSD ideas from the slides:

- technologies are **not neutral**; values can be expressed and embedded in them;
- values and design choices made by some people can have real effects on others, including indirect effects;
- we shape technologies, and technologies shape us;
- actual use depends on stakeholder goals, so intended use is never the whole story;
- unintended uses, non-use, and downstream appropriation matter;
- a design team may decide **not to build** or not to intervene.

Tripartite structure

Investigation type	Focus	Typical question
Conceptual	identify values, stakeholders, trade-offs, and ethical framing	What values are implicated, how should they be understood, and what trade-offs are likely?
Technical	analyse how system features, models, and infrastructures support or undermine values	Which technical choices enable, hinder, or foreclose particular values?
Empirical	engage stakeholders and observe real practices, priorities, and interpretations	How do stakeholders understand, prioritise, and experience the values at stake?

Seven-step VSD process

1. Start with a **value**, a **technology**, or a **context of use**.
2. Identify **direct and indirect stakeholders**.
3. Identify **benefits and harms** for each stakeholder group.
4. Map benefits and harms onto the **corresponding values**.
5. Conduct an investigation of **key values**.
6. Identify **potential value conflicts**.
7. Integrate value considerations into the **organisational structure / development process**.

Useful value vocabulary from the slides

- efficiency
- security
- identity
- calmness
- informed consent
- equality
- privacy
- trust
- accountability
- freedom from harm
- freedom from bias
- well-being
- justice
- autonomy
- safety
- sustainability
- democracy
- transparency
- explainability

Value tensions and designer values

Important VSD additions from the later lecture:

- value analysis should explore **project values**, **designer values**, and **stakeholder values**;
- tensions may arise between these three levels;
- tensions should be acknowledged and mitigated rather than hidden;
- one suggested method is a **value workshop** involving designers and stakeholders;
- values may be universal at a high level but **culturally specific in practice**.

From abstract values to concrete design

The slides explicitly warn against leaving values at a purely abstract level. Following van den Hoven, value discourse should be **operationalised** or **functionally decomposed** into more detailed design implications, sometimes called **moral specs**.

How to operationalise a value
Example:

- **Abstract value**: fairness
- **Operational question**: fair for whom, and against what baseline?
- **Design / evaluation implication**: test false positive and false negative disparities across relevant groups; review annotation instructions; add an appeal or review step.

VSD applied to AI / ML

The slides make three especially important points here:

- AI systems can be **opaque**, so transparency, explainability, and accountability become especially important.
- AI systems can be **adaptive**, so values may drift as the system changes or is repurposed.
- For AI / ML, VSD should extend across the **entire life cycle**, not just the initial design phase.

Transparency, documentation, and accountability

Why documentation matters

Documentation is presented in the slides as more than bureaucracy. It is a method for surfacing assumptions, limits, intended use, values, and possible harms.

- It helps people understand how results might generalise.
- It helps determine how software may be appropriately deployed.
- It helps reveal what biases may be reflected in systems built on a dataset or model.
- It supports transparency, accountability, and better communication to stakeholders.

Data Statements

Data Statements (Bender and Friedman, 2018) are presented in the slides as a documentation practice that draws on VSD and aims to improve engineering and scientific outcomes while enabling more ethically responsive NLP technology.

The schema highlighted in the slides includes:

- curation rationale
- language variety
- speaker demographics
- annotator demographics
- speech situation / context
- text characteristics (genre, topic)
- recording quality
- other relevant information

Other documentation practices you can name

- Data Statements
- Datasheets for Datasets
- Dataset Nutrition Labels
- FactSheets
- Data Cards
- Google Data Cards playbook
- Model Cards for Model Reporting
- System Cards
- Nutrition Labels for Data and Models
- Algorithmic Transparency Reporting Standard

Questions to ask when critiquing documentation

- What is the curation rationale?
- Who is represented, and who is not?
- How were labels created, by whom, and with what instructions?
- What assumptions about language, context, harm, and intended use are visible?

- What deployment limitations or non-recommended uses are disclosed?
- Is subgroup performance discussed?
- Are capabilities and limitations communicated clearly enough for downstream users?

Benefits and barriers

The transparency slides push you to discuss both:

- **benefits**: traceability, accountability, more informed deployment, easier auditing, better communication, easier identification of limitations;
- **barriers**: legal restrictions, commercial secrecy, practical workload, patchy adoption, inconsistent quality, and the possibility that documentation becomes symbolic rather than transformative.

Critical edge to mention
The slides explicitly raise the worry of **"fairness toolkits, a checkbox culture?"** and note that documentation practice remains **ad-hoc and inconsistent**. This is a strong reminder that transparency mechanisms are useful, but not sufficient on their own.

Audits and evaluation

What an algorithmic audit is

An algorithmic audit is a **systematic analysis** of an algorithm or set of algorithms to identify potential biases, errors, or other issues. The slides stress that audits can be conducted by internal or external parties and may still be valuable even without full system access.

Goals of an audit

- evaluate system performance with respect to fairness, equality, and ethics;
- identify limitations and areas for improvement;
- encourage accountability and concrete corrective action.

How to design an audit

1. define the **scope** of the audit;
2. specify the **biases or harms** to investigate;
3. develop a **methodology**;
4. identify the **criteria / metrics**;
5. define the **testing and evaluation process**;
6. establish **benchmarks and performance standards**;
7. document findings, communicate limits, and recommend actions.

Internal vs external audits

Type	Strengths	Limitations
External audit	independent pressure, public scrutiny, can trigger organisational response	less access to internal data, infrastructure, and design rationale
Internal audit	richer access, can happen before release, can create internal processes of responsibility	may face conflicts of interest or weaker external accountability

Important audit cautions from the slides

- audits should be treated as **regular health checks**, not one-off events;
- careful scoping of metrics and datasets is essential;
- you must communicate what an audit **can and cannot demonstrate**;
- benchmarks and metrics are increasingly important in AI governance, but they are not neutral.

From evidence to recommendations

Recommendation categories that fit this module well

Recommendation type	Example action	What value / harm it targets
Data / sampling	rebalance, expand, or resample data; improve subgroup coverage	representation bias, fairness, inclusion
Annotation process	rewrite guidelines, add context, improve annotator training, diversify annotator pool	annotation bias, linguistic fairness, accountability
Model / algorithm	add constraints, tune thresholds, compare subgroup error rates, use counterfactual testing	differential performance, fairness trade-offs
Documentation	publish data statements, model cards, limits, intended use, known failure modes	transparency, responsible deployment, accountability
Audit / monitoring	run regular subgroup audits; track drift; review deployment outcomes	ongoing harm detection, accountability, life-cycle governance

Recommendation type	Example action	What value / harm it targets
Participation	involve affected communities and stakeholders in review or co-design	legitimacy, inclusion, value alignment
Human oversight	add appeals, review processes, escalation paths, or task restrictions	due process, safety, reduction of automation overreach

What makes a recommendation strong

A recommendation should usually specify:

- **what** should change;
- **why** it should change;
- **which stakeholders** benefit or are protected;
- **which value(s)** it supports;
- **which bias / harm** it addresses;
- **what trade-off** it may introduce;
- **how success** would be evaluated.

Recommendation templates you can adapt

Template 1: data / annotation
"Because the audit suggests **annotation or representation bias**, the project should **revise the labelling protocol / expand subgroup coverage / document annotator context**. This would better protect **stakeholder group X** by supporting **fairness, accountability, and freedom from harm**."

Template 2: governance / transparency
"Because users and downstream deployers cannot responsibly interpret the system without better context, the project should publish **data-level and model-level documentation** that states **curation rationale, known limitations, subgroup performance, and non-recommended uses**."

Template 3: stakeholder-centred
"Given the value tension between **A** and **B**, the most defensible recommendation is **Y**, because it reduces harm to **stakeholder group X** without ignoring **stakeholder group Z**. However, it should be paired with **monitoring / review / appeal** because the trade-off cannot be fully eliminated."

A fast workflow for your own ethical problem decisions

Eight-step workflow

1. **State the ethical problem precisely.** What exactly is the concern: unfair flagging, under-protection, opacity, poor documentation, unsafe deployment, exploitative labour, or something else?
2. **Name the stakeholders.** Separate direct and indirect groups, then justify why each matters.
3. **Identify the relevant values.** Use the module vocabulary: fairness, privacy, accountability, safety, autonomy, well-being, transparency, justice, etc.
4. **Diagnose the likely bias mechanism.** Historical, representation, measurement, annotation, aggregation, evaluation, deployment.
5. **Choose fairness / audit measures carefully.** Explain why your chosen metric fits the problem and what it misses.
6. **Critique the documentation.** Ask what a responsible user would need to know but cannot currently know.
7. **Write targeted recommendations.** Tie each one to a value, stakeholder, and implementation path.
8. **Acknowledge limits.** Explain what your evidence does not show, what trade-offs remain, and what should be monitored next.

Compact glossary of high-value terms

Term	Short assignment-use definition
Algorithmic bias	systematic unfair discrimination or differential treatment in computational systems
Protected characteristic	socially significant attribute linked to anti-discrimination concerns
Stakeholder	anyone significantly implicated by a system, directly or indirectly
Direct stakeholder	interacts with the system or its outputs
Indirect stakeholder	affected without direct interaction
Value Sensitive Design (VSD)	design approach that systematically incorporates human values into technology design
Conceptual investigation	clarifies values, stakeholders, ethical framing, trade-offs
Technical investigation	studies how infrastructures, models, and design choices support or undermine values
Empirical investigation	studies how stakeholders actually experience, prioritise, and interpret values

Term	Short assignment-use definition
Value tension	conflict between values, or between stakeholder, designer, and project priorities
Designer values	personal and professional values that designers bring into the process
Moral specs	concrete design implications derived from abstract values
Representation bias	under-representation of part of the use population
Measurement bias	bad feature / label choices, often driven by poor proxies
Annotation bias	systematic distortion introduced by subjective labelling processes
Aggregation bias	use of one model for heterogeneous groups where subgroup differences matter
Evaluation bias	mismatch between benchmark / test data and real use population
Deployment bias	mismatch between intended task and real-world institutional use
Procedural fairness	fairness of process
Outcome fairness	fairness of resulting outcomes or distributions
Demographic parity	equal positive rates across groups
Equalized odds / equal opportunity	equal true-positive and/or false-positive rates across groups
Calibration	same score has same meaning across groups
Fairness through unawareness	flawed idea that removing protected attributes alone yields fairness
Algorithmic audit	systematic evaluation of an AI system for biases, errors, harms, and limitations
Data Statement	documentation practice for surfacing context, assumptions, and values in NLP datasets
Datasheet for Datasets	structured dataset documentation for transparency and responsible reuse
Model Card	model-level documentation covering intended use, factors, metrics, and limits
System Card	system-level transparency document describing capabilities, constraints, and risks

Final takeaways to remember in your presentation

The big ideas most worth carrying into the assignment

1. **Technology is not neutral.**
2. **Fairness is plural and contested, not singular.**
3. **Different stakeholders care about different errors and harms.**
4. **Bias can enter at many points: data, labels, model, evaluation, deployment.**
5. **VSD is useful because it links values, stakeholders, and design choices.**
6. **Transparency and documentation help, but they are not magic fixes.**
7. **Audits should be systematic, scoped carefully, and repeated over time.**
8. **Good recommendations are specific, evidence-based, value-aware, and implementable.**

End of guide.